

# Assignment 1: Linux User Management & Automation

## 1. Introduction/Objective-

### Objective:

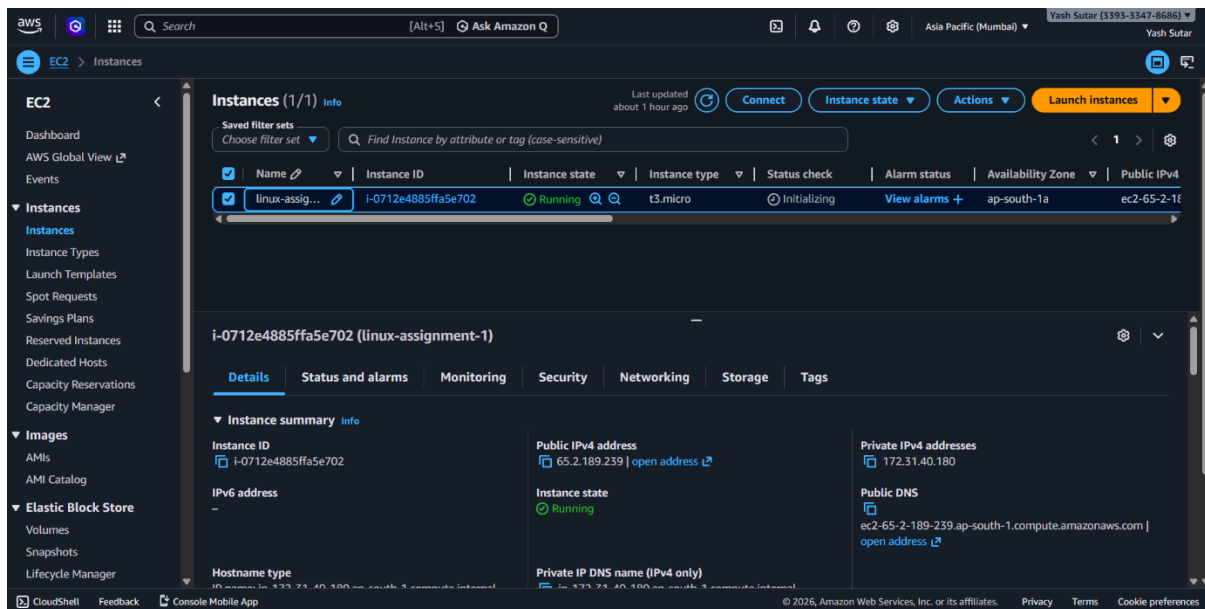
To set up and manage user access on an Ubuntu EC2 instance by implementing role-based permissions, secure SSH key-based authentication, and automation using shell scripting and cron jobs for system maintenance.

## 2. AWS EC2 Setup-

### Content:

An Ubuntu EC2 instance was launched on AWS. The instance was accessed securely using an SSH key pair from a local machine. This instance served as the environment where all user management, security configurations, and automation tasks were performed.

Fig. 1: This screenshot shows the running Ubuntu EC2 instance used for the assignment.



### 3. User Creation-

#### Content:

Three users were created to represent different roles in a system:

- admin-user → Full administrative access
- dev-user → Limited access for development tasks
- readonly-user → Read-only access with no system privileges

This approach ensures role-based access control, which is essential for system security and management.

Fig. 2: Admin user: This screenshot shows the creation of the admin-user using the adduser command.

```
ubuntu@ip-172-31-40-180:~$ sudo adduser admin-user
info: Adding user 'admin-user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group 'admin-user' (1001) ...
info: Adding new user 'admin-user' (1001) with group 'admin-user (1001)' ...
info: Creating home directory '/home/admin-user' ...
info: Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for admin-user
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] Y
info: Adding new user 'admin-user' to supplemental / extra groups 'users' ...
info: Adding user 'admin-user' to group 'users' ...
ubuntu@ip-172-31-40-180:~$ |
```

Fig. 3: Dev user & Readonly user: This screenshot shows the creation of dev-user and readonly-user accounts.

```
ubuntu@ip-172-31-40-180:~$ sudo adduser dev-user
info: Adding user `dev-user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `dev-user' (1002) ...
info: Adding new user `dev-user' (1002) with group `dev-user (1002)' ...
info: Creating home directory `/home/dev-user' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for dev-user
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n]
info: Adding new user `dev-user' to supplemental / extra groups `users' ...
info: Adding user `dev-user' to group `users' ...
ubuntu@ip-172-31-40-180:~$
sudo adduser readonly-user
info: Adding user `readonly-user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `readonly-user' (1003) ...
info: Adding new user `readonly-user' (1003) with group `readonly-user (1003)' ...
info: Creating home directory `/home/readonly-user' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for readonly-user
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `readonly-user' to supplemental / extra groups `users' ...
info: Adding user `readonly-user' to group `users' ...
```

## 4. Sudo Permission Configuration-

### Content:

User permissions were configured using the visudo file to control access:

- admin-user → Full sudo access
- dev-user → Limited sudo access (only apt commands)
- readonly-user → No sudo access

This ensures that only authorized users can perform critical system operations.

### Why full sudo access is dangerous:

Giving full sudo access to all users removes security boundaries and allows any user to modify or damage the system.

Fig. 4: Sudo Configuration using visudo

This screenshot shows the configuration that restricts dev-user to specific commands.

```
# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
dev-user ALL=(ALL) /usr/bin/apt
```

Fig. 5: **dev-user Limited Sudo Access**

This screenshot demonstrates that dev-user can execute apt commands but is restricted from other administrative commands.

```
ubuntu@ip-172-31-40-180:~$ su - dev-user
Password:
dev-user@ip-172-31-40-180:~$ sudo apt update
[sudo] password for dev-user:
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
dev-user@ip-172-31-40-180:~$ sudo reboot
Sorry, user dev-user is not allowed to execute '/usr/sbin/reboot' as root on ip-172-31-40-180.ap-south-1.compute.internal.
dev-user@ip-172-31-40-180:~$ |
```

Fig. 6: **readonly-user Sudo Restriction**

This screenshot shows that readonly-user is denied sudo access.

```
dev-user@ip-172-31-40-180:~$ su - readonly-user
Password:
readonly-user@ip-172-31-40-180:~$ sudo apt update
[sudo] password for readonly-user:
readonly-user is not in the sudoers file.
readonly-user@ip-172-31-40-180:~$ |
```

## 5. SSH Key-Based Authentication-

### Content:

SSH key-based authentication was implemented to provide secure access to the system.

- A key pair was generated on the local machine
- The public key was added to each user's authorized\_keys file
- Users were able to log in without passwords

### Concept:

- Public key → Stored on server
- Private key → Kept secure on local machine

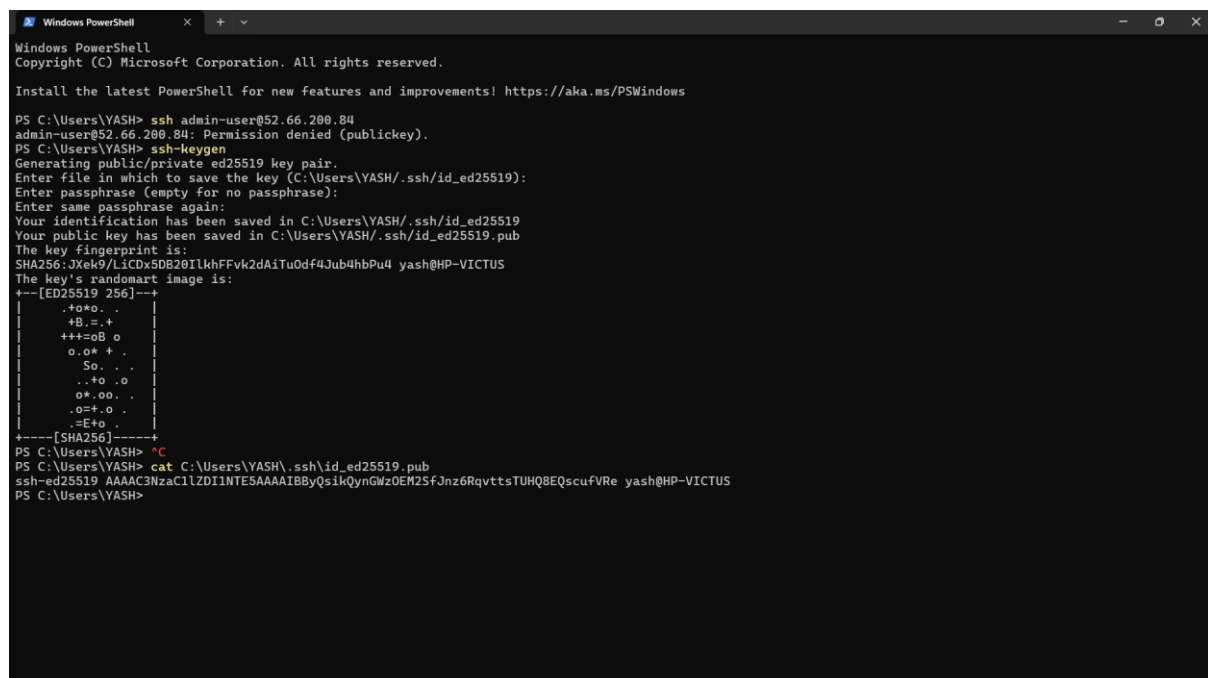
This method is more secure than password-based authentication.

### Challenges encountered:

Initially, a fingerprint (SHA256) was confused with the public key, which caused SSH authentication failures. This was resolved by correctly using the public key.

Fig. 7: SSH Key Generation

This screenshot shows the generation of SSH keys on the local machine.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\YASH> ssh admin-user@52.66.200.84
admin-user@52.66.200.84: Permission denied (publickey).
PS C:\Users\YASH> ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\YASH/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\YASH/.ssh/id_ed25519
Your public key has been saved in C:\Users\YASH/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:JXek9/LiCDxSDB20IlkhFFvk2dAiTuOdF4Jub4hbPu4 yash@HP-VICTUS
The key's randomart image is:
+--[ED25519 256]--+
| .+o*o. . |
| +B.+. |
| +++=oB o |
| o.o* + . |
| So. . . |
| ..+o .o |
| o*.oo. . |
| .o=+.o . |
| .E+o . |
+----[SHA256]-----+
PS C:\Users\YASH> ^C
PS C:\Users\YASH> cat C:\Users\YASH/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBByQsikQynGwzOEM2SfJnz6RqvttstUHQ8EQscufVRe yash@HP-VICTUS
PS C:\Users\YASH>
```

Fig. 8: `authorized_keys` Configuration

This screenshot shows the public key added to the `authorized_keys` file for users.

```
ubuntu@ip-172-31-40-180:~$ sudo cat /home/admin-user/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBByQsikQynGWzOEM2SfJnz6RqvttstUHQ8EQscufVRe yash@HP-VICTUS
ubuntu@ip-172-31-40-180:~$ sudo cat /home/dev-user/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBByQsikQynGWzOEM2SfJnz6RqvttstUHQ8EQscufVRe yash@HP-VICTUS
ubuntu@ip-172-31-40-180:~$ |

ubuntu@ip-172-31-40-180:~$ sudo mkdir -p /home/readonly-user/.ssh
ubuntu@ip-172-31-40-180:~$ sudo nano /home/readonly-user/.ssh/authorized_keys
ubuntu@ip-172-31-40-180:~$ sudo chown -R dev-user:dev-user /home/dev-user/.ssh
sudo chmod 700 /home/dev-user/.ssh
sudo chmod 600 /home/dev-user/.ssh/authorized_keys
ubuntu@ip-172-31-40-180:~$ sudo cat /home/readonly-user/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBByQsikQynGWzOEM2SfJnz6RqvttstUHQ8EQscufVRe yash@HP-VICTUS
ubuntu@ip-172-31-40-180:~$ |
```

Fig. 9: SSH Login as `admin-user`

This screenshot shows successful password-less login using SSH key authentication.

```
PS C:\Users\YASH> ssh admin-user@52.66.200.84
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr  4 11:19:51 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   33.9% of 6.71GB Processes:    116
Memory usage: 26%          Users logged in: 1
Swap usage:   0%           IPv4 address for ens5: 172.31.40.180

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

admin-user@ip-172-31-40-180:~$
admin-user@ip-172-31-40-180:~$ |
```

Fig. 10: SSH Login as dev-user

```
PS C:\Users\YASH> ssh dev-user@52.66.200.84
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr  4 11:34:43 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   34.1% of 6.71GB Processes:      114
Memory usage: 25%          Users logged in: 1
Swap usage:   0%           IPv4 address for ens5: 172.31.40.180

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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applicable law.

dev-user@ip-172-31-40-180:~$ |
```

Fig. 11: SSH Login as readonly-user

```
PS C:\Users\YASH> ssh readonly-user@52.66.200.84
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Apr  4 11:38:47 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   34.2% of 6.71GB Processes:      116
Memory usage: 25%          Users logged in: 1
Swap usage:   0%           IPv4 address for ens5: 172.31.40.180

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

readonly-user@ip-172-31-40-180:~$ |
```



## 6. Bash Script Automation-

### Content:

A script named `system-update.sh` was created to automate system updates. The script updates packages, upgrades installed software, and logs the output with timestamps.

This reduces manual effort and ensures consistent system maintenance.

### Script:

```
#!/bin/bash

echo "System update started at $(date)" >> update.log

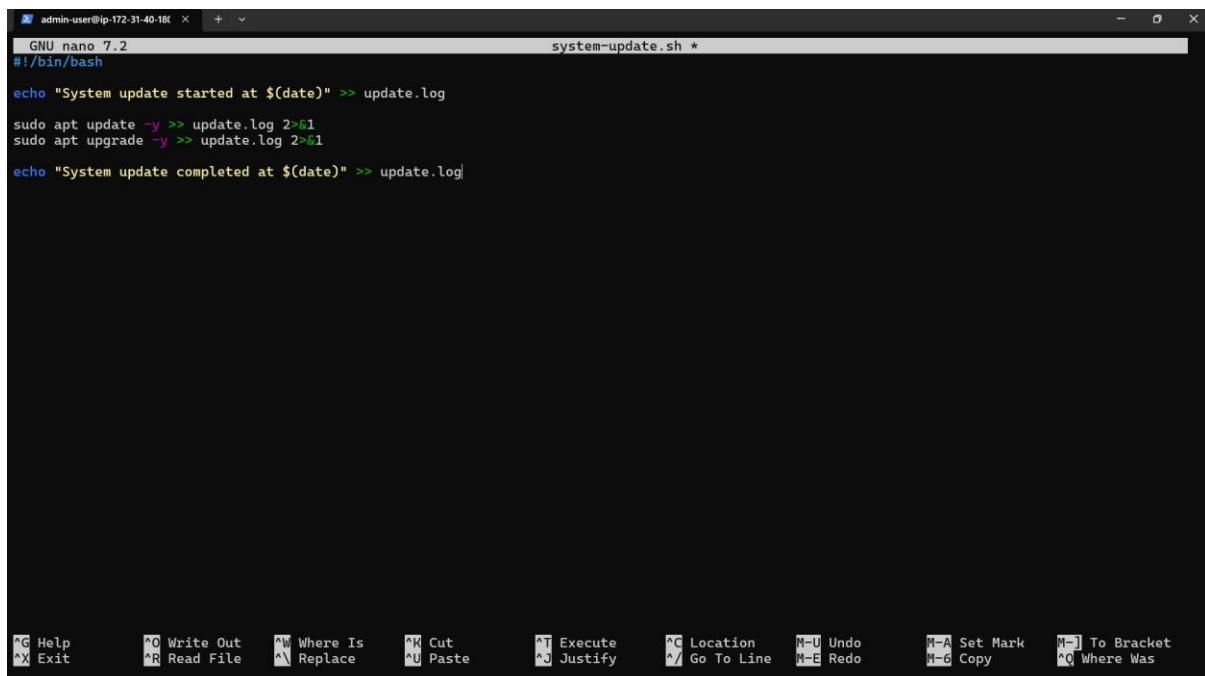
sudo apt update -y >> update.log 2>&1

sudo apt upgrade -y >> update.log 2>&1

echo "System update completed at $(date)" >> update.log
```

**The script automates system updates and stores output in a log file, which helps in monitoring system activity and debugging errors.**

Fig. 12: This screenshot shows the script created using the nano editor.

A screenshot of a terminal window showing the nano text editor editing a file named 'system-update.sh'. The editor's title bar indicates 'GNU nano 7.2' and the file name. The script content is as follows:

```
#!/bin/bash

echo "System update started at $(date)" >> update.log

sudo apt update -y >> update.log 2>&1
sudo apt upgrade -y >> update.log 2>&1

echo "System update completed at $(date)" >> update.log
```

The bottom of the screen displays a row of keyboard shortcuts for nano, such as 'G Help', 'O Write Out', 'W Where Is', 'C Cut', 'T Execute', 'G Location', 'U Undo', 'A Set Mark', 'I To Bracket', 'X Exit', 'R Read File', 'A Replace', 'U Paste', 'J Justify', 'G Go To Line', 'E Redo', 'C Copy', and 'Q Where Was'.

## 7. Logging-

### Content:

The script logs all outputs into a file (update.log). Logging helps track system activities and identify issues during execution.

### Fig. 13: Log Output

This screenshot shows the log file generated after running the script.

```
ubuntu@ip-172-31-40-180:~$ su - admin-user
Password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

admin-user@ip-172-31-40-180:~$ nano system-update.sh
admin-user@ip-172-31-40-180:~$ chmod +x system-update.sh
admin-user@ip-172-31-40-180:~$ ./system-update.sh
[sudo] password for admin-user:
admin-user@ip-172-31-40-180:~$ cat update.log
System update started at Sat Apr  4 11:59:31 UTC 2026

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists...
Building dependency tree...
Reading state information...
All packages are up to date.

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Reading package lists...
Building dependency tree...
Reading state information...
Calculating upgrade...
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
System update completed at Sat Apr  4 11:59:38 UTC 2026
admin-user@ip-172-31-40-180:~$ |
```

## 8. Cron Job Scheduling-

### Content:

The script was scheduled using cron to run automatically every day at midnight.

Cron entry:

```
0 0 * * * /home/admin-user/system-update.sh
```

### Explanation:

0 → minute

0 → hour (midnight)

\* → every day

\* → every month

\* → every week

This ensures automated system updates without manual intervention.

Fig. 14: Cron Job Configuration

```
admin-user@ip-172-31-40-180:~$ crontab -e
no crontab for admin-user - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano          <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]: |
```

Fig. 15: Cron Job Verification

```
admin-user@ip-172-31-40-180:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 0 * * * /home/admin-user/system-update.sh
admin-user@ip-172-31-40-180:~$ |
```

## **9. Challenges Encountered**

### **Content:**

During the assignment, the following challenges were faced:

- Confusion between public key and fingerprint
- Permission denied (publickey) error during SSH login
- Incorrect file naming instead of `authorized_keys`
- Understanding sudo restrictions for different users
- Managing file permissions for SSH authentication

These challenges were resolved through troubleshooting and better understanding of Linux concepts.

## **10. Conclusion-**

### **Content:**

This assignment provided hands-on experience with Linux system administration concepts such as user management, access control, SSH authentication, scripting, and task automation. It also strengthened understanding of AWS EC2 and real-world DevOps practices.